# Analytical Modelling of NR-V2X Sidelink Groupcast and Relay Selection/Placement over Platoon Mobility Traces

Companion analytical study to the `relay_groupcast_eval` ns-3 campaign

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**Abstract**

We develop a closed-form analytical model of the NR sidelink (Mode 2) groupcast system implemented in the `csi/relay_groupcast_eval` ns-3 simulator, and use it to explore relay selection and placement *before* committing to simulation. The model expresses (i) the resource pool as a function of its numerology/bitmap/PSFCH configuration; (ii) a groupcast transmission with its HARQ/PSFCH feedback budget; (iii) multi-flow coexistence under sensing-based semi-persistent scheduling (SB-SPS) and a deterministic (collision-free) baseline; (iv) per-receiver PRR, PIR and goodput; (v) resource *requirement* as a function of traffic and resource *usage* as a function of channel condition; and (vi) a two-hop decode-and-forward relay with several selection/placement variants. A four-parameter link abstraction is calibrated to the ns-3 PRR-versus-distance curves over the actual platoon geometry, reaching an RMSE of 0.030. All metrics are then evaluated over two SUMO traces (an 8-vehicle inter-vehicle-distance sweep and an $8 \to 23$ incremental merge) in MATLAB. The analysis predicts that a single mid-platoon relay placed at $x^\star \approx 20$–$40\,\mathrm{m}$ (merge) raises group PRR by up to $+0.16$ for high MCS, that the relay link itself should use a *robust* modulation, and that returns from additional relays saturate after the second hop.

## 1 System model and notation

A single leader (node 0) periodically multicasts an 800 B application payload every 10 ms (CAM-like, 0.64 Mbps) to all platoon followers on an NR-SL groupcast bearer. The physical layer uses numerology $\mu$, a sub-channelised resource pool with HARQ feedback over PSFCH, and a 3GPP V2V highway channel with geometric NLOSv blockage in which every vehicle is a potential blocker. Table 1 lists the configuration as model variables; these mirror the simulator one-to-one.

## 2 Resource-pool model

With $1000 \cdot 2^\mu$ slots per second, a fraction $\rho$ usable for sidelink and a fraction $1/N_{\mathrm{psfch}}$ reserved for feedback, the pool offers a PSSCH capacity (in sub-channel·slots per second)

$$R_{\mathrm{pool}} \;=\; \underbrace{1000 \cdot 2^{\mu}}_{\text{slots/s}} \cdot \rho \;\cdot\; \left(1 - \tfrac{1}{N_{\mathrm{psfch}}}\right) \;\cdot\; N_{\mathrm{sub}} \;=\; 11\,250\,\text{subch}\cdot\text{slot/s}. \tag{1}$$

Table 1: Resource-pool, traffic and HARQ parameters (model variables).

| Symbol | Meaning | Value |
|---|---|---|
| $\mu$ | numerology | 1 (30 kHz SCS) |
| $T_{\text{slot}}$ | slot duration | 0.5 ms |
| $N_{\text{sub}}$ | number of sub-channels | 10 |
| $W$ | sub-channel size | 10 PRB |
| $\rho$ | usable-slot fraction (bitmap `111111000111`) | 9/12 |
| $N_{\text{psfch}}$ | PSFCH period | 4 slots |
| $g_{\min}$ | `MinTimeGapPsfch` | 3 slots |
| $N_{\max}$ | max transmissions per TB | 2 |
| $B$ | application payload | 800 B |
| $\Lambda$ | packet rate | 100 pkt/s |
| RRI, PDB | reservation interval / delay budget | 10 ms |

A transport block of $B' = (B + \text{hdr})$ bytes at MCS spectral efficiency $\eta_m$ needs

$$\text{PRB}_m = \left\lceil \frac{8B'}{\eta_m R_{\text{RE}}} \right\rceil, \qquad c_m = \left\lceil \frac{\text{PRB}_m}{W} \right\rceil \text{ sub-channels}, \qquad \sigma_m = \left\lceil \frac{c_m}{N_{\text{sub}}} \right\rceil \text{ slots}, \tag{2}$$

with $R_{\text{RE}}$ the data resource elements per PRB. Equation (2) reproduces the simulator's grant sizing ($c_8 = 6$, $c_{28} = 2$ sub-channels) and exposes the low-MCS *resource-starvation* regime: MCS 0 needs $c_0 = 26$ sub-channels, i.e. $\sigma_0 = 3$ slots, occupying CR = 38% of the pool (Fig. 2).

## 3 Groupcast transmission and HARQ/PSFCH feedback

A groupcast TB to a receiver experiencing block-error rate BLER is delivered, under chase-combining HARQ with up to $N$ copies, with probability

$$\text{PRR} \;=\; 1 - \prod_{c=1}^{N} \text{BLER}\big(\gamma + (c-1)\Delta_{\text{cc}}\big), \tag{3}$$

where $\gamma$ is the per-copy SINR and $\Delta_{\text{cc}}$ the combining gain. The number of feedback-driven retransmissions that fit inside the delay budget is bounded by the HARQ round-trip time, itself set by the PSFCH period and minimum gap:

$$\tau_{\text{harq}} \approx \left( \left\lceil \frac{g_{\min}}{N_{\text{psfch}}} \right\rceil + 1 \right) N_{\text{psfch}}\, T_{\text{slot}} \;\approx\; 4\,\text{ms}, \qquad N \le \min\left( N_{\max},\; \left\lfloor \frac{\text{PDB}}{\tau_{\text{harq}}} \right\rfloor + 1 \right). \tag{4}$$

With PDB = 10 ms and $N_{\text{psfch}} = 4$, (4) admits at most one feedback-driven retransmission ($N = 2$), exactly the saturation observed in simulation: redundancy is capped by the PDB and PSFCH timing, *not* by the RRI.

## 4 Multi-flow coexistence: SB-SPS vs deterministic scheduling

Under SB-SPS, each transmitter reserves resources within a selection window of $N_{\text{csr}} = N_{\text{sub}} \cdot \lceil W_{\text{sel}}\, \rho \rceil$ candidate single-slot resources. For $m$ mutually-sensing transmitters the probability that a given grant collides is

$$p_{\text{col}}(m) \;=\; 1 - \left(1 - \tfrac{1}{N_{\text{csr}}}\right)^{m-1}, \tag{5}$$

which is zero for a single transmitter (the leader-only groupcast) and grows linearly for small $m$ (Fig. 2, bottom-right). A deterministic (pre-assigned) scheduler sets $p_{\text{col}} = 0$ by construction and upper-bounds achievable reliability; the gap between the two quantifies the cost of contention once a relay adds a second transmitter. The aggregate demand of a set of flows $\mathcal{F}$ is

$$D = \sum_{f \in \mathcal{F}} \Lambda_f \, \mathbb{E}[N_{\text{tx},f}] \, c_{m_f}, \qquad \text{CR} = D / R_{\text{pool}}, \tag{6}$$

so the channel-occupancy ratio is the natural congestion metric tying traffic and MCS to the pool of (1).

## 5 Channel/link model and calibration

We summarise distance- and blockage-dependent link quality by an effective excess loss and a per-MCS logistic decode probability:

$$g(d, n) = \underbrace{\alpha \log_{10} d}_{\text{path loss}} + \underbrace{\beta \, n}_{\text{NLOSv}}, \qquad \text{PRR}_m(d, n) = \tfrac{1}{2} \operatorname{erfc}\!\Big( \tfrac{g(d,n) - \theta_m}{\sqrt{2}\, s_m} \Big), \tag{7}$$

where $d$ is the TX–RX distance, $n$ the number of vehicles geometrically obstructing the link (counted per the simulator's blocker test), $\theta_m$ the per-MCS loss margin and $s_m$ its transition width. Equation (7) is the reduced form of the SINR/BLER chain of §3 ($\theta_m = \text{SNR}_0 - \gamma_m$). Its parameters are fitted to the ns-3 PRR($d$) curves *over the actual trace geometry*: a 2-D search over $(\alpha, \beta)$ with an inner per-MCS fit of $(\theta_m, s_m)$. The calibration (Fig. 1) attains RMSE = 0.030 with $\alpha = 8.0$, $\beta \approx 0$: in a single-lane platoon, blocker count is strongly correlated with range, so distance alone is a sufficient statistic for link quality, and the two traces collapse onto one curve per MCS. The fitted $\theta_m$ decreases with MCS index (less loss tolerated), and MCS 0 sits far below the rest — its low PRR is resource starvation, not decoding, consistent with §2.

## 6 PRR, PIR and throughput

For a receiver at $(d, n)$ the per-link reliability is $\text{PRR}_m(d, n)$ from (7). For periodic traffic at interval RRI with independent losses, the mean packet inter-reception time and application goodput are

$$\text{PIR}_m(d, n) = \frac{\text{RRI}}{\text{PRR}_m(d, n)}, \qquad S_m(d, n) = \Lambda \, B \, \text{PRR}_m(d, n). \tag{8}$$

Evaluated over the IVD geometry (Fig. 3), robust MCS hold PRR $> 0.95$ and PIR $\approx$ RRI across the platoon, while high MCS inflate PIR and depress goodput in the tail. The *group* PRR is the membership-weighted mean over all active receivers, $\text{GPRR}(t) = \frac{1}{|\mathcal{R}(t)|} \sum_{r \in \mathcal{R}(t)} \text{PRR}(d_r(t), n_r(t))$.

## 7 Resource requirement vs traffic, and usage vs channel

Two complementary views follow from the model. *Requirement* is set by traffic and MCS via (2)–(6): at fixed payload, CR falls from 38% (MCS 0) to $\sim$2% (MCS 28), so the choice of modulation is the dominant lever on channel load. *Usage* is modulated by the channel through the expected number of transmissions $\mathbb{E}[N_{\text{tx}}] = 1 + (N_{\max} - 1)\,\overline{\text{BLER}}$, which is U-shaped in MCS (Fig. 2, bottom-left): minimal ($\approx$1.01) for mid MCS, rising at MCS 0 (starvation, 1.64) and at MCS 28 (decode failures, 1.13). The product $\mathbb{E}[N_{\text{tx}}]\, c_m$ is the effective per-packet footprint that a relay must be weighed against.

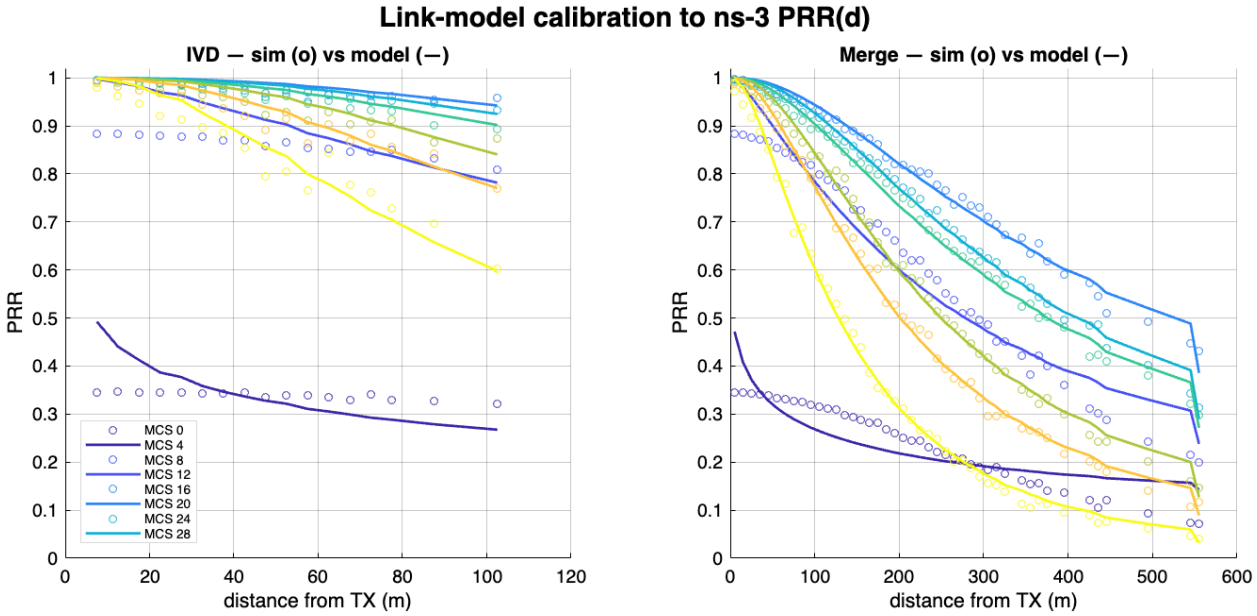



Figure 1: Reduced-form link model (7) (lines) versus ns-3 simulated PRR (circles) for both traces and all MCS. RMSE = 0.030.

# 8 Relay selection and placement

## 8.1 Two-hop decode-and-forward with cooperative union

A relay placed near a boundary distance $x$ from the leader decodes the leader's packet and re-broadcasts it on a second groupcast link (Broadcast cast type, no PSFCH, so no NACK storm). A far receiver $f$ is served if *either* the direct or the relayed copy decodes:

$$\mathrm{PRR}_f^{\mathrm{union}} = 1 - \big(1 - \mathrm{PRR}_{L\to f}\big)\big(1 - \mathrm{PRR}_f^{\mathrm{relay}}\big), \quad \mathrm{PRR}_f^{\mathrm{relay}} = \mathrm{PRR}_{L\to r} \cdot \mathrm{PRR}_{r\to f} \cdot \mathbb{1}[d_{Lr} < d_{Lf}], \tag{9}$$

where $r$ is the elected relay and the indicator enforces that the relay is ahead of $f$. The two terms expose the placement tension: a relay close to the leader ($x$ small) decodes almost surely ($\mathrm{PRR}_{L\to r} \to 1$) and contributes a *diversity* copy, but its hop to $f$ is nearly as long as the direct link; a relay further back shortens the second hop (*range extension*) but is itself less reliable. The optimum $x^\star$ balances the two.

## 8.2 Selection variants

We model: (i) *strict-closest* selection, electing per packet the active node whose live distance is nearest the boundary $x$ (the policy implemented in the simulator); (ii) *best-link* selection, electing the node maximising $\mathrm{PRR}_{L\to r} \cdot \overline{\mathrm{PRR}_{r\to f}}$; (iii) a *multi-relay chain* with relays at $\{10, 20, 50, 100\}$ m and union over all hops; and (iv) a *relay-MCS* degree of freedom, letting the (shorter, less-obstructed) relay hop use a different modulation than the leader.

## 8.3 Results

Figure 4 sweeps the relay boundary $x$. For the merge trace the gain peaks sharply at $x^\star \approx 20$–$40$ m and is largest for high MCS (up to +0.16 GPRR for MCS 28, recovering most of the tail it loses without a relay); placing the relay too far back ($x > 150$ m) collapses the gain because the relay

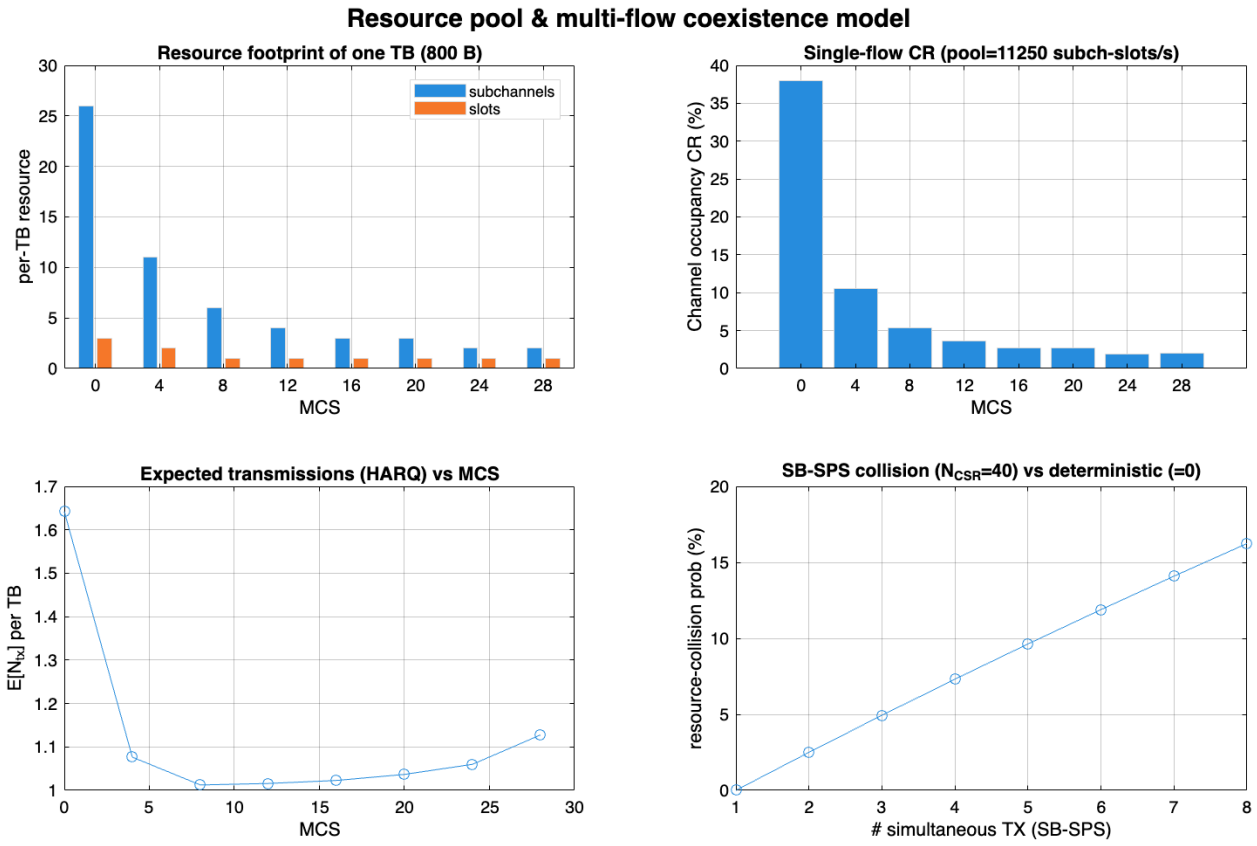



Figure 2: Resource-pool/coexistence model: per-TB footprint (2), single-flow CR (6), expected transmissions (3), and SB-SPS collision (5) versus a deterministic (0) baseline.

can no longer decode the leader. For the IVD trace, distances are short and the platoon already well-served, so the benefit is a smaller diversity gain maximised at small $x$ (Table 2). Figure 5 shows that (a) a chain saturates after the second relay (diminishing union returns at fixed geometry); (b) the relay link is best operated at a *robust* MCS (≈MCS 8) even when the leader uses MCS 20 — the short relay hop does not need a high rate and benefits from the reliability; and (c) strict-closest selection matches best-link to within ∼0.5 percentage points, so the simple, low-overhead geometric policy is near-optimal here.

Table 2: Analytical relay-placement optimum (cooperative union, strict-closest).

| Trace | MCS | no-relay GPRR | $x^\star$ (m) | best GPRR ($\Delta$) |
|---|---|---|---|---|
| IVD | 20 | 0.963 | 5 | 0.998 (+0.034) |
| IVD | 28 | 0.872 | 15 | 0.977 (+0.104) |
| Merge | 20 | 0.827 | 40 | 0.929 (+0.102) |
| Merge | 24 | 0.780 | 30 | 0.904 (+0.124) |
| Merge | 28 | 0.668 | 20 | 0.828 (+0.160) |

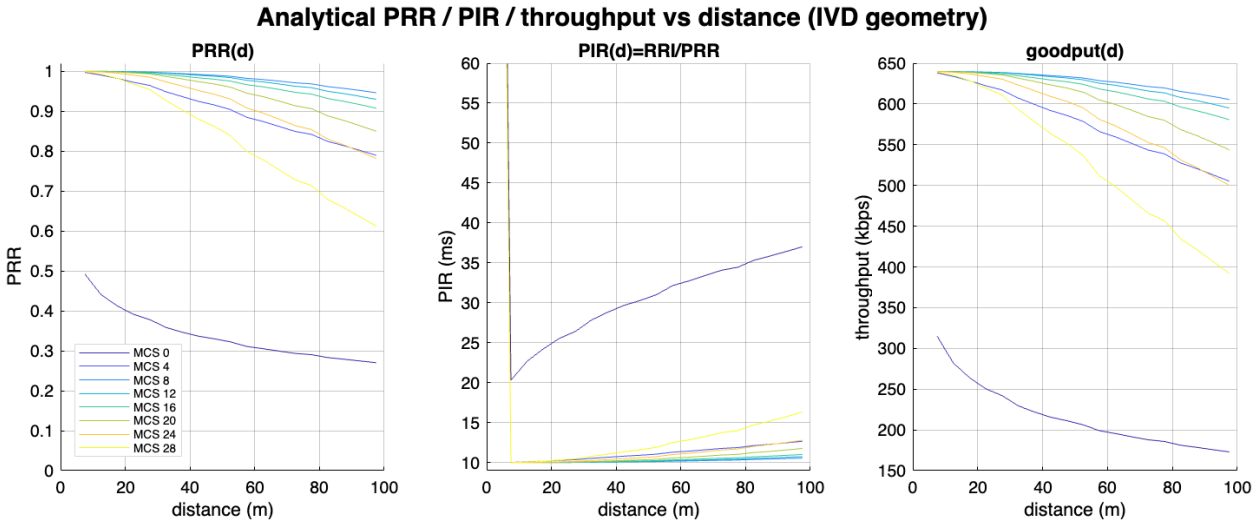



Figure 3: Analytical PRR, PIR and goodput versus distance per MCS (8) (IVD geometry). MCS 0 is the resource-starved floor.

# 9 Conclusions and guidance for simulation

The calibrated model (RMSE 0.030) reproduces the simulated link behaviour and lets us scan placements and variants cheaply. Its predictions, to be confirmed by the `relay_groupcast_eval` campaign, are: (1) a single relay is worth deploying mainly for *high* MCS and *long* platoons (merge), where it recovers 0.10–0.16 of tail GPRR; (2) the optimal relay sits mid-platoon at $x^\star \approx 20$–$40\,\mathrm{m}$, not at the tail; (3) the relay hop should use a robust modulation regardless of the leader's MCS; (4) strict-closest geometric selection is near-optimal, justifying the simple in-simulator policy; and (5) a second relay adds little once the first is placed well, so a chain is rarely justified under this QoS. These narrow the simulation grid to the high-MCS×mid-$x$ region and motivate adding a robust-relay-MCS arm.

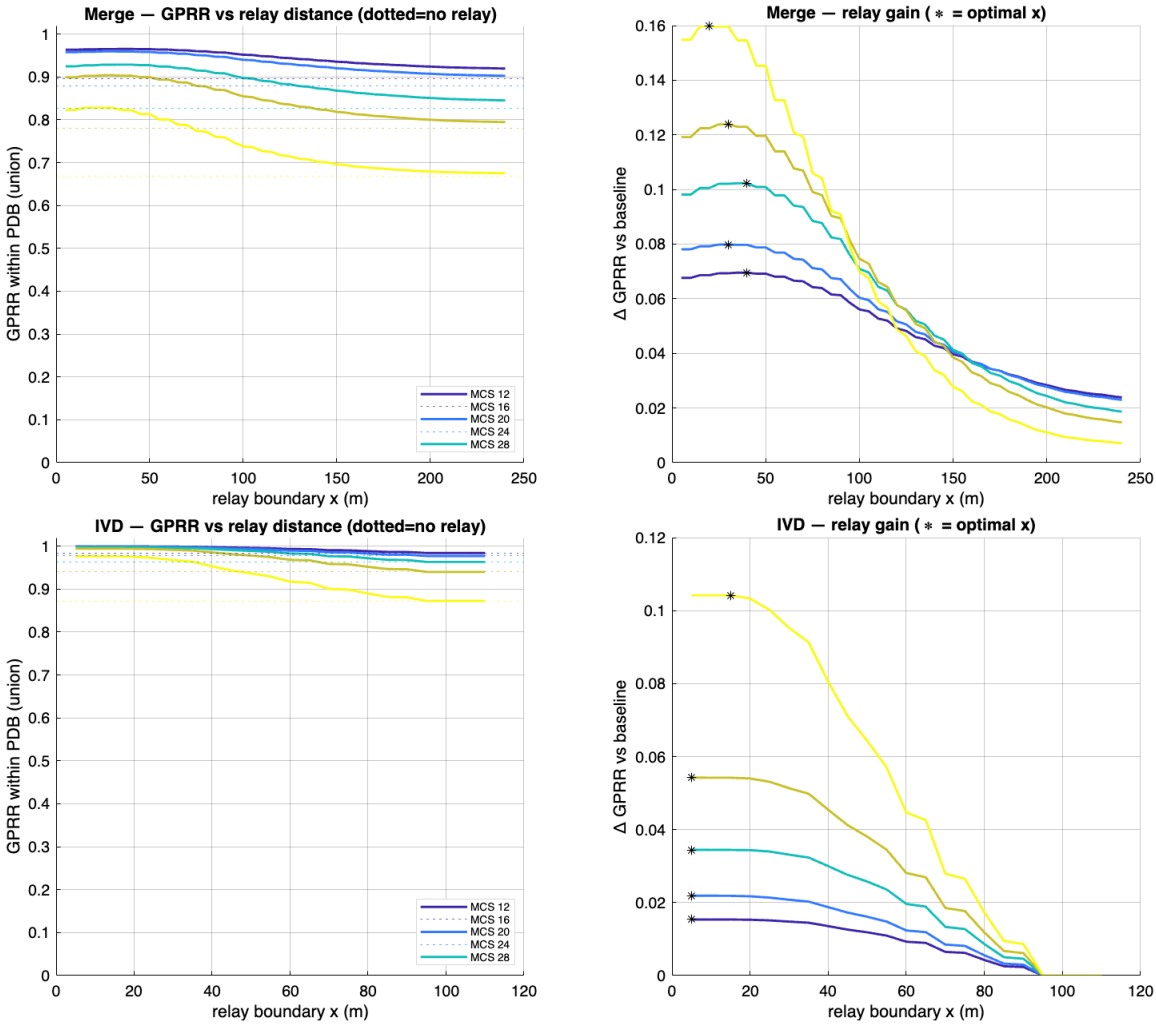



Figure 4: GPRR (within PDB, union) versus relay boundary $x$ and the resulting gain over the no-relay baseline (dotted), per MCS. Top: merge; bottom: IVD. $*$ marks the optimal placement.

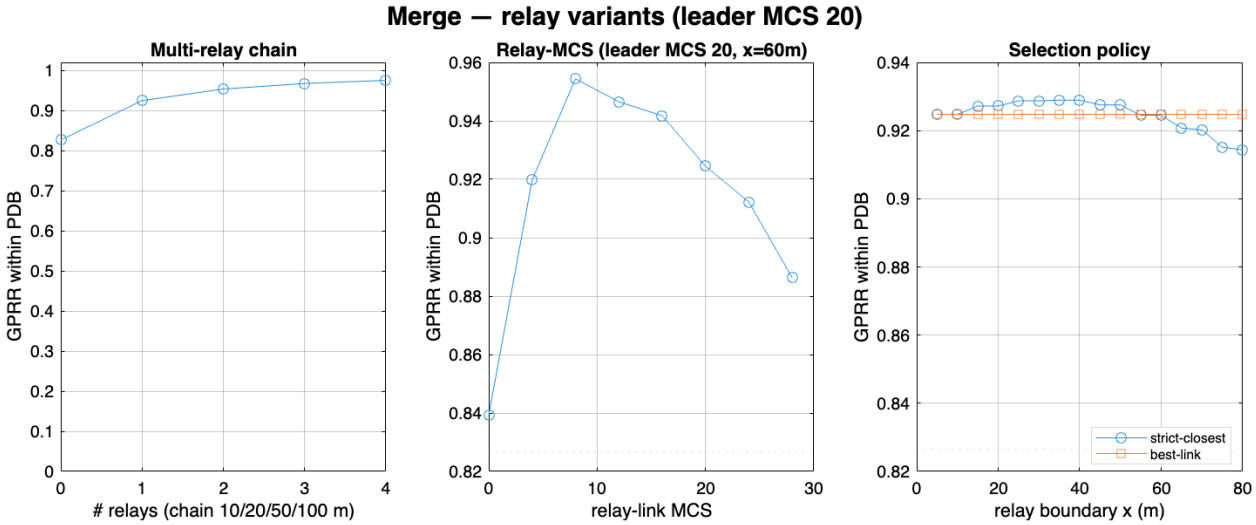



Figure 5: Relay variants (merge, leader MCS 20): multi-relay chain saturation; relay-link MCS sweep (optimum at a robust MCS); strict-closest vs best-link selection.

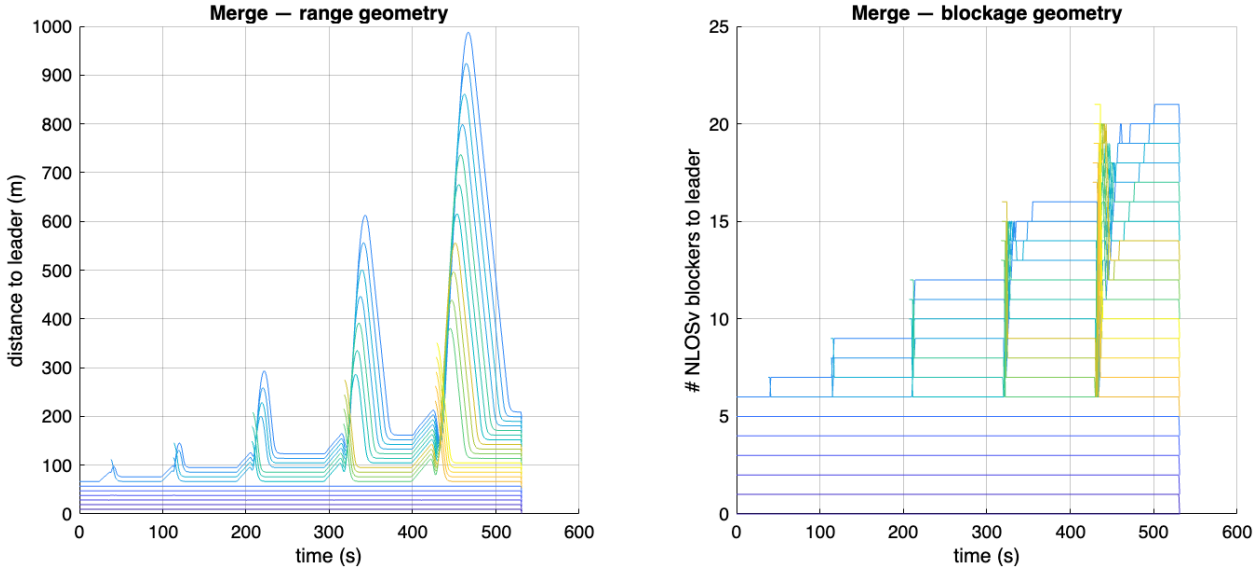



Figure 6: Merge trace geometry driving the model: per-follower range and NLOSv blocker count to the leader over time (junctions add vehicles in steps).